

C101



SEAT No. _____

SARDAR PATEL UNIVERSITY

M.Sc. (IT) SEMESTER - 1

PS01CINT53: COMPUTER NETWORKS & CYBER SECURITY

Monday, 14th February, 2022

Max. Marks: 70

Time: 12:00 pm to 2:00 pm

Q-1

[8]

(A) Choose the most appropriate option for each question:

1. _____ of OSI model is responsible for synchronization.
(A) Physical layer (B) Data link layer
(C) Session layer (D) Presentation layer
2. The _____ satellites are so close to the earth, the ground stations do not need much power, and the round-trip delay is least.
(A) LEO (B) MEO
(C) GEO (D) None of these
3. Mobile is an example of _____ transmission mode.
(A) Simplex (B) Half-duplex
(C) Full-duplex (D) None of these
4. In Gigabit Ethernet the speed of communication is measured in _____.
(A) Mbps (B) Gbps
(C) Kbps (D) None of these
5. Conversion of plain text into cipher text is known as _____.
(A) Encoding (B) Decoding
(C) Encryption (D) Decryption
6. In _____ attack, an attacker only read and interpret the message passing between sender and receiver.
(A) Passive (B) Normal
(C) Active (D) None of these
7. Process of transferring an ownership of mobile device from one base station (tower) to another base station is known as _____.
(A) Cut off (B) Hand off
(C) Switch off (D) Shut down
8. Full form of VPN is _____.
(A) Virtual Public Network (B) Virtual Private Network
(C) Virtual Private Node (D) Virtual Public Node

[16]

(B) Do as directed.

1. TCP/IP model has 7 layers. True / False
2. Twisted pair is an unguided media. True / False
3. An area of the Earth covered by the microwave radiation from a satellite transponder is known as footprint. True / False
4. Router device is work only up to data link layer. True / False
5. Routing algorithm at the network layer is responsible for deciding which output line an incoming packet should be transmitted on. True / False
6. Radio waves are transmitted in all the directions. True / False
7. Dynamic routing algorithms change their routing decisions to reflect changes in the topology and traffic. True / False
8. To connect isolated hosts and networks using other networks is known as tunneling. True / False

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9. In symmetric key algorithm sender and receiver uses the same key.

True / False.

10. In context of fragmentation, MTU stands for _____.
11. When too many packets are present at the router, performance degrades. This situation is called _____.
12. Keeping information secret from intermediate users is known as _____.
13. Send the same message indiscriminately to a large number of internet users is known as _____.
14. In _____ attack, machine is slowdown or shut down and making it inaccessible to its intended users.
15. In mobile communication, AMPS stands for _____.
16. Full form of MTSO, in context of mobile communication _____.

Q-2 Answer the following questions (ANY SEVEN):

[14]

1. Difference between Hub and Switch.
2. Full form of OSI and TCP/IP
3. Difference between fiber and satellite.
4. Explain flooding in brief.
5. List different ways in which networks differ.
6. Difference between LAN & MAN.
7. Define: Orbit and Microcell
8. Difference between Secret-key and Public-key algorithms.
9. Define: Firewall & Cybercrime

Q-3 Explain OSI reference model in detail.

[8]

OR

Q-3 List guided and unguided medias. Explain any two in brief.

[8]

Q-4 Write a short note on communication satellites.

[8]

OR

Q-4 List static and dynamic routing algorithms. Explain any one in detail.

[8]

Q-5 Define: fragmentation. List types of fragmentation. Explain any one type in detail.

[8]

OR

Q-5 Explain mobile communication in brief.

[8]

Q-6 Using transposition cipher method convert following plaintext into cipher text with key = "APPLE".

[8]

"GOLDEN STATUE IS IN ELEVENTH CAVE"

OR

Q-6 Explain Phishing, Key loggers, Trojan horse & Denial of service.

[8]

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